

NovaStar MX (COEX)

SIMPL+ Control Module

User Manual & Integration Guide

Release 2.20.1

Crestron 4-Series • NovaStar COEX TCP/IP API

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1. Introduction

NovaStar MX (COEX) SIMPL+ Module for Crestron 4-Series

The **NovaStar MX SIMPL+ Module** provides direct TCP/IP control of NovaStar LED processors (COEX platform, MX30 and compatible) from a Crestron 4-Series control system. Communication is handled over the standard NovaStar COEX API on **TCP port 8001**.

Typical use cases include:

- Recalling named presets stored on the NovaStar controller
- Adjusting LED wall brightness via an analog signal (0–100)
- Muting and unmuting the picture output
- Querying and displaying the current display mode
- Showing live preset names in a Crestron touch panel UI

This module targets Crestron 4-Series processors only. It is not compatible with 3-Series hardware.

2. System Requirements

Component	Requirement
Crestron Processor	4-Series (CP4, MC4 or equivalent)
Crestron Firmware	v2.0 or newer
SIMPL Windows	Version 4.17.00 or later
TCP Network Access	Port 8001 open between processor and NovaStar unit
NovaStar Hardware	MX30 controller (COEX platform)
NovaStar Firmware	Compatible with the COEX TCP/IP API

Ensure that any firewalls or VLAN ACLs between the Crestron processor and the NovaStar controller permit outbound TCP connections on port 8001.

3. Installation Guide

3.1 Import the Module

1. Open your project in SIMPL Windows 4.17.00 or later.
2. Go to **File** → **Import** and select the **.usp** (SIMPL+) source file provided in the package.
3. Alternatively, place the pre-compiled **.clz** file in your project's module directory so SIMPL Windows picks it up automatically.
4. The module will appear in the Symbol Library under the imported modules category.

3.2 Place the Symbol

Drag the NovaStar MX symbol onto your SIMPL program sheet. Connect the required input and output signals as described in Section 4.

3.3 TCP Client Configuration

The module manages its own TCP client connection internally. You do **not** need to add a separate TCP/IP Client symbol. Simply drive the following signals at startup:

Controller_IP\$ → e.g. "192.168.1.100"

Controller_Port → 8001 (default – leave at 0 to use the built-in default)

3.4 Compile & Upload

1. Compile the SIMPL program (F5).
2. Upload to the Crestron processor via Toolbox or SIMPL Windows.
3. Verify the module connects by checking the **DisplayMode_FB\$** signal — it should return a non-empty string within a few seconds of the processor booting.

4. Signal Reference

4.1 TCP Settings

Signal	Type	Description
Controller_IP\$	String	IP address of the NovaStar controller
Controller_Port	Analog	TCP port for API communication (default: 8001)

4.2 Control Inputs

Signal	Type	Description
Get_Display_Mode	Digital	Pulse HIGH to request the current display mode from the controller.
Picture_Mute	Digital	Pulse HIGH to mute the LED wall output.
Picture_UnMute	Digital	Pulse HIGH to unmute the LED wall output.
Get_Presets	Digital	Pulse HIGH to fetch all preset names from the controller.
Set_Brightness	Analog	Drive 0–100 to set the LED wall brightness. Changes take effect immediately.
Set_Preset	Analog	Drive 1–10 to recall the corresponding preset on the controller.

4.3 Feedback Outputs

Signal	Type	Description
DisplayMode_FB\$	String	Current display mode as a human-readable string.
Brightness_FB	Analog	Live brightness value (0–100) reflected from the controller.
PresetActive	Analog	The number (1–10) of the currently active preset.
PresetName\$[1–10]	String	Array of preset name strings returned by the controller (1-indexed).

5. Configuration & Usage

5.1 Initial Connection

Set **Controller_IP\$** to the NovaStar unit's IP address at system startup (e.g. from a string parameter or a persistent string signal). The module will establish the TCP connection automatically and begin heartbeat polling.

5.2 Recalling Presets

Drive the **Set_Preset** analog signal with a value of 1–10 to recall a preset. The **PresetActive** feedback will update to reflect the new active preset once the controller confirms the change.

*Call **Get_Presets** at startup to populate the **PresetName\$[1–10]** strings so your touch panel can display human-readable preset names.*

5.3 Brightness Control

Connect a touch panel slider (0–100) or any analog source to **Set_Brightness**. The current value is reflected in **Brightness_FB** for UI feedback.

5.4 Picture Mute

Pulse **Picture_Mute** to blank the LED wall output. Pulse **Picture_UnMute** to restore it. These map directly to the corresponding COEX API commands.

5.5 Automatic Reconnection

The module includes a built-in heartbeat and recovery mechanism. If the TCP connection drops (e.g. due to a network interruption or controller reboot), the module will attempt to reconnect automatically — no intervention required.

6. Troubleshooting

No feedback received after boot

- Verify the IP address in Controller_IP\$ is correct.
- Confirm TCP port 8001 is reachable from the processor (use Toolbox TCP test).
- Check for firewall or VLAN rules blocking port 8001.
- Ensure the NovaStar MX30 controller is powered on and connected to the network.

PresetName\$ strings are empty

- Pulse Get_Presets to request the preset list from the controller.
- The controller must have presets configured before names are available.

Set_Preset has no effect

- Ensure the value driven is in the range 1–10.
- Confirm the preset exists on the controller.
- Check that the TCP connection is established (DisplayMode_FB\$ should be non-empty).

Module not found in SIMPL Windows

- Verify the .usp or .clz file is in the correct module directory.
- Restart SIMPL Windows after placing the file.
- Check the SIMPL Windows version is 4.17.00 or later.

7. Release Notes

Version	Date	Changes
2.20.1	April 2026	Packaging cleanup. Removed unused SIMPL# references and XML documentation files. Re
2.20	January 2026	Stability update for MX30 state handling. Improved heartbeat monitoring and connection reco
2.1	November 2025	Initial public release of the NovaStar MX COEX SIMPL+ module.

All releases within the 2.x line are backward-compatible. Upgrading from an earlier 2.x version requires only replacing the module file and recompiling.

8. Support & Contact

For integration assistance, custom Crestron development, or questions about this module, contact **SAOA Consulting**:

Website	https://saoa.se
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Scope	Crestron integration, NovaStar LED control, AV system design

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